



UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Advanced Level

ECONOMICS

Paper 3 Multiple Choice (Supplement)

9708/03

May/June 2009

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

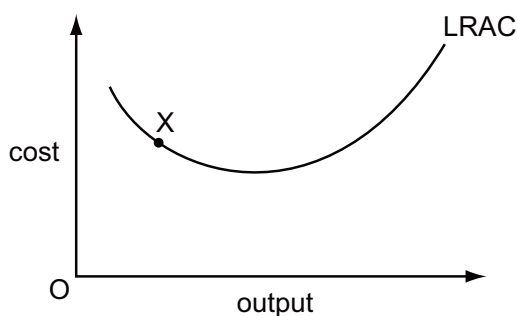
Zarak Mushtaq

Solved with reasons

This document consists of **12** printed pages.



- 1 In the diagram, the firm is operating at point X on its long-run average cost curve.



LONG-RUN AVERAGE COST: The per unit cost of producing a good or service in the long run when all inputs under the control of the firm are variable. In other words, long-run total cost divided by the quantity of output produced. Long-run average cost is guided by returns to scale.

Which statement about the firm is correct?

- A It is operating at its optimal level of output. *LRAC = min*
- ☒ B It is operating below its cost-minimising level of output. *(MES)*
- C It is productively inefficient.
- D It could produce its current level of output at a lower cost.
- } Efficient at that particular plant*
- 2 The table shows the marginal utility derived by a consumer who devotes the whole of his weekly income of \$42 to two goods X and Y, whose unit prices are \$3 and \$6 respectively.

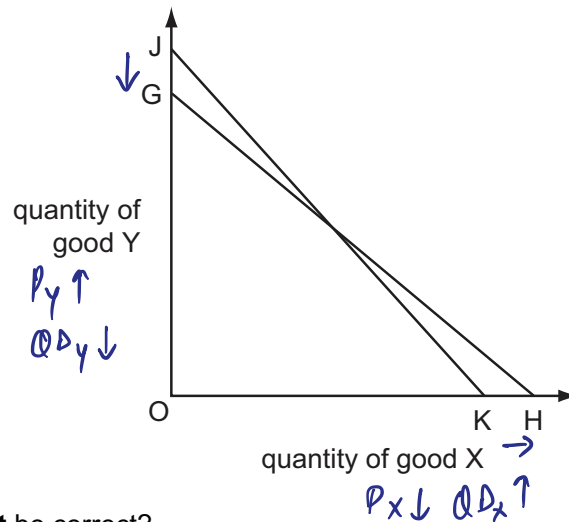
unit	marginal utility of X (units)	marginal utility of Y (units)
1	12 <i>4</i>	34 <i>5.6</i>
2	11 <i>3.7</i>	30 <i>5</i>
3	10 <i>3.3</i>	26 <i>4.3</i>
4	9 <i>3</i>	22 <i>3.6</i>
5	8 <i>2.6</i>	18 <i>3</i>
6	7 <i>2.3</i>	14 <i>2.3</i>
7	6 <i>2</i>	10 <i>1.6</i>
8	5 <i>1.6</i>	6 <i>1</i>

In order to maximise his utility, which quantities of X and Y should the consumer purchase?

	X $\frac{MU_X}{P_X}$	Y $\frac{MU_Y}{P_Y}$
☒ A	2 <i>3.7</i>	6 <i>2.3</i>
☒ B	4 <i>3</i>	5 <i>3</i>
C	6 <i>2.3</i>	4 <i>3.6</i>
D	8 <i>1.6</i>	3 <i>4.3</i>

$$\frac{MU_X}{P_X} = \frac{MU_Y}{P_Y} \text{ (Equi-marginal principle)}$$

- 3 In the diagram a consumer's budget line shifts from JK to GH.



Which statement **must** be correct?

- A There has been an increase in the consumer's money income.
 B There has been a decrease in the consumer's real income. (shift)
 ✓ C Good Y has become 'relatively' more expensive.
 D The price of good X has increased. (one side change)
- 4 To increase its labour force from 100 to 101 workers, a firm has to increase its daily wage rate from \$400 to \$405.

What is the marginal cost of labour per day?

$$TC_{101} - TC_{100} = MC_L$$

$$(101 \times 405) - (100 \times 400) = 905$$

- A \$5 B \$405 ✓ C \$905 D \$40 905

- 5 An economist calculates that a firm has incurred the following costs over the course of a year.

↓
 Considers
 opportunity
 cost

	\$(000)
wages and salaries	150
<u>opportunity cost of owner's time</u>	40
materials	80
rent	30
marketing fees	20
interest on bank loans	25
<u>interest forgone on finance provided by owner</u>	15

$$40 + 15 = \underline{\underline{\$55000}}$$

By how much does total cost as defined by an economist exceed the total cost as defined by an accountant?

- A \$15 000 B \$40 000 ✓ C \$55 000 D \$85 000

- 6 A manufacturing firm has one plant of optimum size.

The firm builds a second plant identical to its first plant. The firm then finds that its long-run average cost has risen. [DEOS]

What could account for the change in its long-run average cost?

- A diminishing returns (short run concept - Law of variable proportions)
 B external diseconomies of scale (This shifts)
~~C~~ managerial diseconomies of scale (communication, long chain of command)
 D technical diseconomies of scale (Nothing like that)

- 7 A firm estimates that, all else remaining unchanged, an increase in its output will result in an equal proportionate increase in its revenue. Qty ↑ TR ↑, same % , price unchanged

What can be deduced from this about the price elasticity of demand for the firm's product?

- A It is -1. (unitary - TR unchanged)
 B It is +1. (Doesn't exist)
 C It is perfectly inelastic. (Qty unchanged)
~~D~~ It is perfectly elastic. (price unchanged)
↳ Apply concept of perfect competition

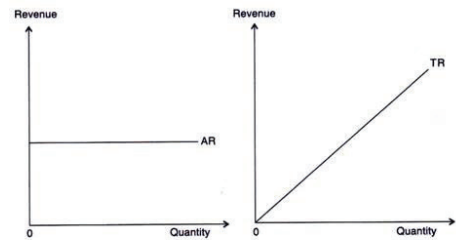
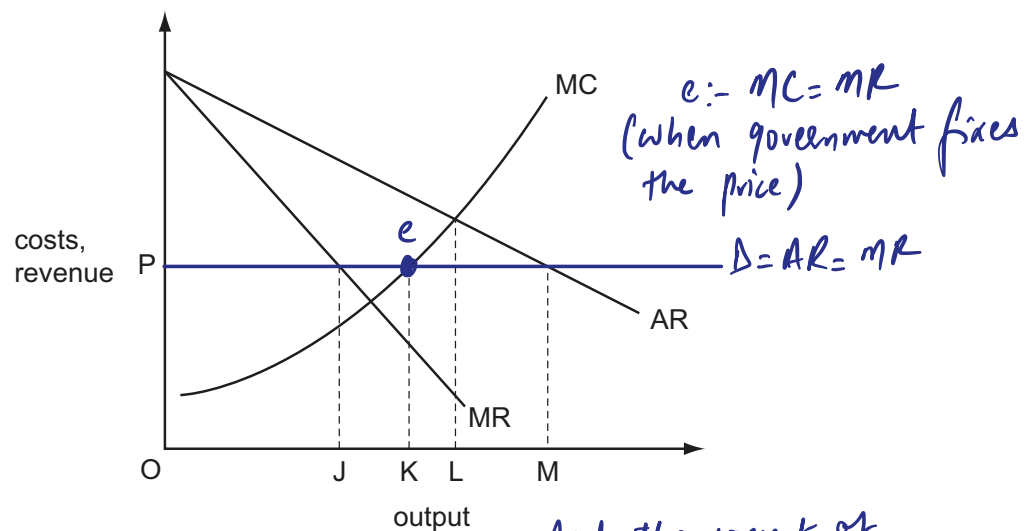


Fig. 1 The average and total revenue curves of a perfectly competitive firm

- 8 The diagram shows the initial cost and revenue curves of a profit-maximising monopolist.



What output will the firm produce if the government fixes the price at OP?

- A OJ ~~B~~ OK C OL D OM

Apply the concept of perfect competition
 ↓
 MR = AR = P

Imperfect competition / Existence of advertising

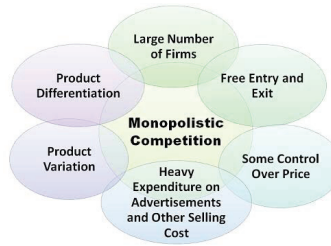


5

→ Perfect knowledge

9 What is a feature of monopolistic competition, but not of perfect competition?

- A a small number of buyers
- ☒ B product differentiation
- C the existence of abnormal profits
- D the existence of barriers to entry

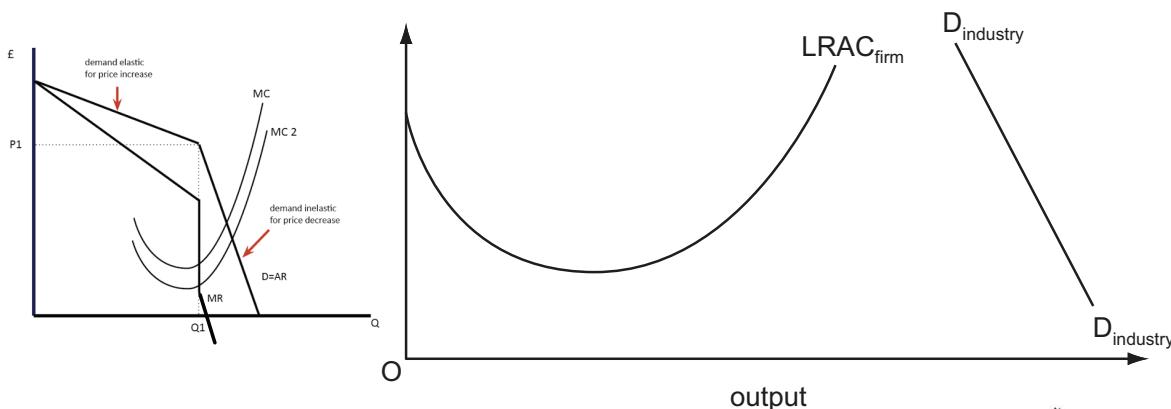


10 Which assumption is essential for a market to be contestable?

- A The market is supplied by a large number of firms.
- ☒ B Firms are free to enter and leave the market.
- C Firms cannot earn abnormal profits in the short run.
- D Firms produce differentiated goods.

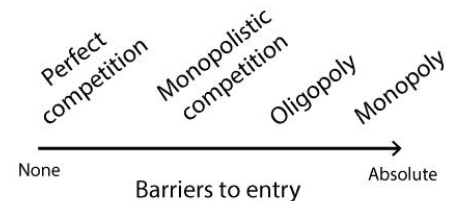
A monopoly being a monopoly can't act as a monopoly because of threat of competition as the cost of entry and exit are zero.

11 The diagram shows the long-run average cost curve of a typical firm in an industry and the demand curve for the industry's product.



Which market structure is most likely to occur in this industry?

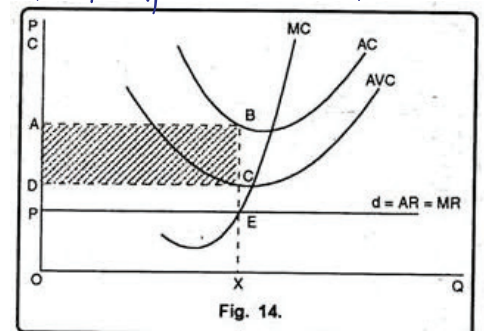
- A monopolistic competition *Elastic demand*
- B monopoly *25%*
- ☒ C oligopoly → *concentration ratio, kink demand curve*
- D perfect competition *Perfectly elastic demand*



12 In which circumstance will a firm cease production in the short run?

- A It makes a profit that is less than its total variable costs.
- B It makes a profit that is less than its total fixed costs.
- C Its average revenue is less than its average cost.
- ☒ D Its average revenue is less than its average variable cost.

→ Apply perfect and monopolistic



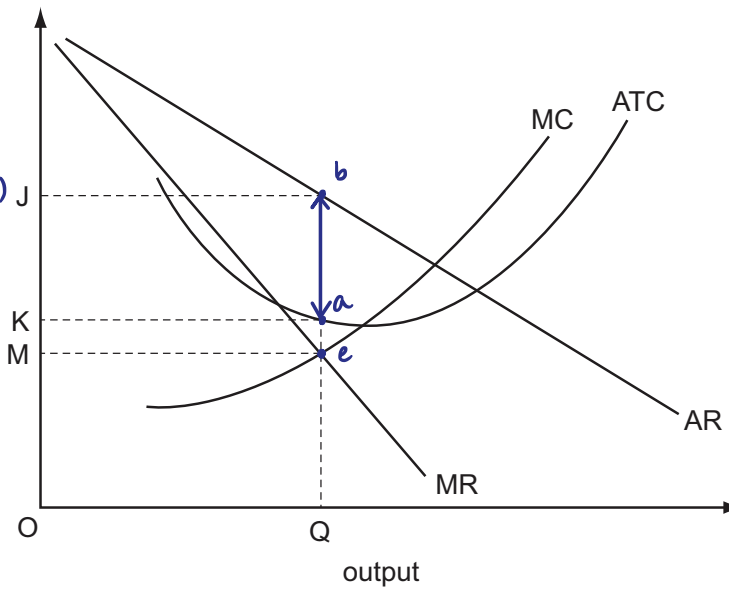
$$MC = MR$$

13 The diagram shows the cost and revenue curves of a profit-maximising monopolist.

e:- $MC = MR$, determines the profit maximizing level of output

a':- cost determination (P) J
revenue, cost

b':- Price determination (C) K
M



What measures the monopoly profit per unit of output made by the firm?

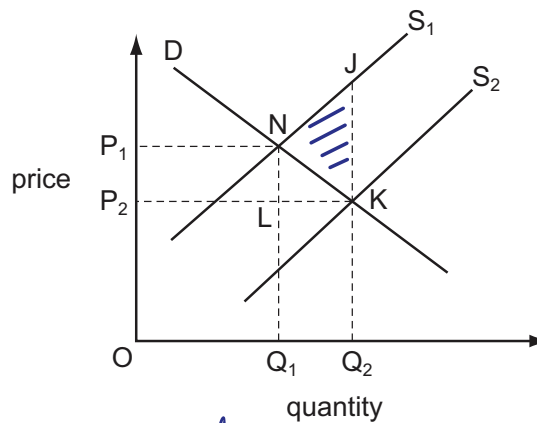
A JM

☒ B JK

C $JM \times OQ$

D $JK \times OQ$

14 In the diagram the introduction of a government subsidy causes an industry's supply curve to shift from S_1 to S_2 .



welfare loss

Which area measures the resulting deadweight loss to society?

A P_1NKP_2

☒ B JKN

C NLK

D Q_1Q_2JN

↓
increase
in consumer
surplus

15 A country's national income per head increases.

S.O.L

What could explain why this is accompanied by a fall in households' standard of living?

- ✓ A an increase in personal taxes *e.g. income tax*
- B an increase in the trade deficit, *More imports, S.O.L ↑*
- C an increase in population, *National income per head ↓*
- D a rise in the exchange rate, *Appreciation makes imports cheap, S.O.L ↑*

$S + T + M$

16 Which of the following correctly identifies net leakages from the circular flow of income?

	<i>injection</i> trade surplus (exports - imports)	<i>injection</i> government budget deficit (government spending - taxes)	<i>leakage</i> private sector surplus (saving - investment)
A	✓	✓	<i>More 'S'</i> X
B	✓	X ✓	X
C	X ✓	<i>Downward</i> ✓	✓ ✓
D	X ✓	X ✓ <i>Vertical</i>	✓ ✓

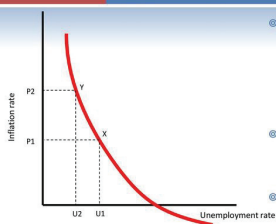
Phillips curve (Short run and long run)

17 According to monetarist theory, which policy objectives are in conflict in the short run, but not in the long run?

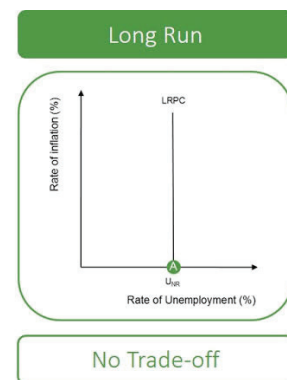
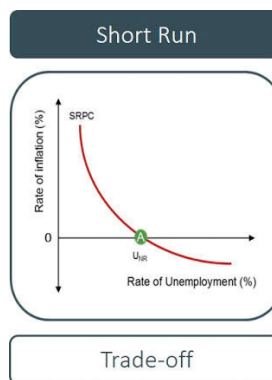
- A economic growth and full employment, *Economic growth, incomes ↑, employment ↑*
- B economic growth and price stability, *output ↑, AS ↑, prices ↓*
- ✓ C price stability and full employment
- D price stability and equilibrium in the balance of payments *Prices ↓, BOP surplus
↳ link with trade*

	Keynesian view	Monetarist view
Fiscal policy	In recession, expansionary fiscal policy can stimulate economic activity	Fiscal policy causes no long-term increase in real output
Wage rigidity	Wages can be sticky downwards causing unemployment	In absence of min wages/ trade unions wages flexible.
Unemployment	Demand-deficient unemployment big causes	Tend to emphasize supply-side unemployment (natural rate)
Phillips Curve	There is a trade off between unemployment and inflation	Only a trade-off in the short-term.
Government borrowing	In recession, governments should borrow more to offset fall in private spending	Government should seek to run balanced budget
Crowding out	No crowding out in recession	Government borrowing causes more crowding out

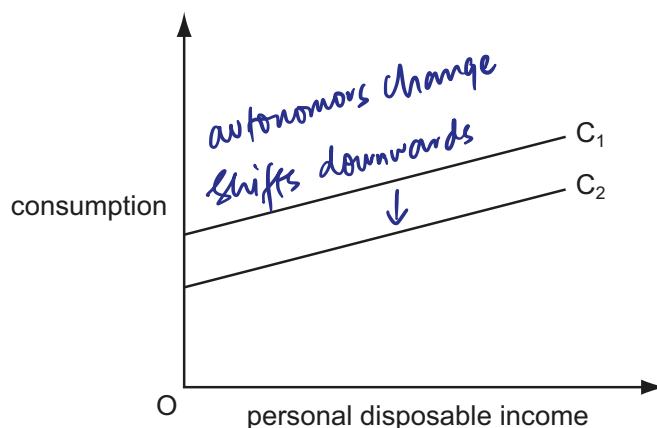
THE SHORT-RUN PHILLIPS CURVE



- Phillips' work suggests that changes in unemployment have direct and, to some degree, predictable effects on inflation, and therefore have implications for economic policy makers e.g. lower inflation is achievable at the expense of a higher rate of inflation
- The Phillips Curve is not therefore a theory of inflation, but illustrates the conflict between two objectives
- For example, assume the economy is at point X, and wishes to reduce unemployment to U2, it may increase government spending and therefore aggregate demand in order to move the economy to point Y. However, this is at the cost of higher inflation at P2 because the pool of unemployed workers will fall, firms will compete for workers by raising wages, thus increasing wages costs, which are passed on in the form of higher prices



18 A country's initial consumption function is C_1 .



What would be most likely to cause the consumption function to shift from C_1 to C_2 ?

- A a decrease in personal disposable income, *written on x-axis, movement*
- B a decrease in the expected future rates of income tax *C↑*
- ☒ C an increase in interest rates *↑ save ↑ C↓*
- D an increase in wealth *C↑*

19 In a closed economy with no government $C = 30 + 0.7Y$, where C is consumption and Y is income.

The equilibrium level of income is 300. *AE = Y*

What is the level of investment?

- ☒ A 60
- B 100
- C 210
- D 270

$$AE = C + I \quad (\text{Remember } AE = Y)$$

$$Y = C + I$$

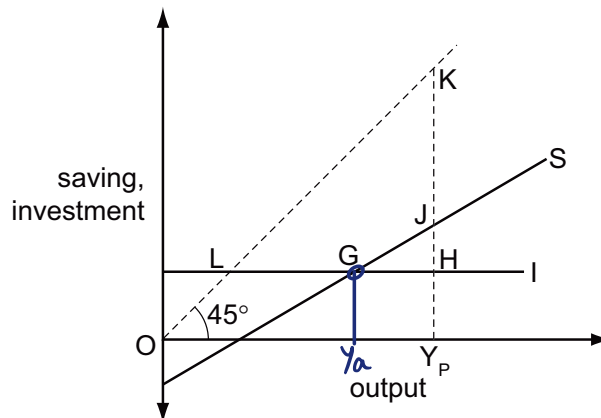
$$300 = 30 + 0.7Y + I$$

$$300 = 30 + 0.7(300) + I$$

$$300 = 240 + I$$

$$I = 60$$

20 The diagram shows the saving and investment curves of a closed economy with no government.



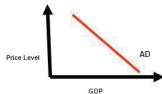
The potential level of output is OY_P .

Which distance measures the gap between actual and potential output?

- A LG ☒ B GH C JH D KJ
- $(Y_a - Y_P)$

21 The diagram shows a country's aggregate demand curve.

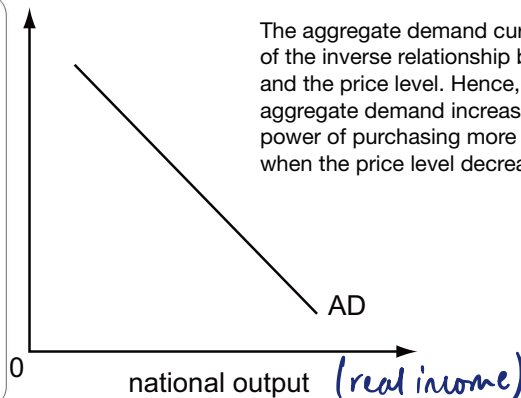
Why is Aggregate Demand Curve Downward Sloping



Explain how each of the following effects helps explain why the AD curve is downward sloping?

Interest Rate Effect

A lower price level decreases the demand for money, which decreases the interest rate and increases investment and interest sensitive components of consumption and, therefore, Real GDP or real output.



The aggregate demand curve is negatively-sloped, because of the inverse relationship between aggregate expenditures and the price level. Hence, if the price level decreases, the aggregate demand increases and vice-versa. Hence, the power of purchasing more goods at the same price, occurs when the price level decreases.

What could explain why the curve slopes downwards?

- ☒ A A fall in the price level increases the real value of money balances.
- B A fall in the price level leads to an increase in interest rates.
- C A fall in the price level leads to a rise in the real exchange rate.
- D A fall in the price level leads to the expectation of a further decrease in the price level.

- 22 According to Keynesian theory, in which circumstance would there always be an increase in the demand for money?

	real income	price level	interest rates
A	increase ✓	decrease	increase
B	constant	constant	increase
☒ C	increase ✓	increase ✓	decrease ✓
D	constant	decrease	decrease ✓

Real income ↑, more demand for T+P+S

Prices ↑, to maintain purchasing power, more demand for money

R ↓, Save ↓, Spend ↑, demand for money ↑ for T+P+S

- 23 How is outward migration from a developing economy likely to affect its balance of payments?

- A It may improve its balance of payments by increasing its export capacity. *No link*
- ☒ B It may improve its balance of payments by increasing inflows of current transfers. *No*
- ☒ C It may worsen its balance of payments by causing a currency depreciation. *No*
- D It may worsen its balance of payments by increasing consumer expenditure on imported goods. *(No link to migration)*

payments received from workers

- 24 The table shows the figures for consumption, gross capital formation and depreciation in four economies, all measured in US \$.

economy	consumption (\$m)	gross capital formation (\$m)	depreciation (\$m)
A	200	40	50
☒ B	500	200	150
C	1 000	1 200	1 400
D	20 000	6 000	6 000

= -10

= +50

= -200

= 0

- 25 Which change would best indicate that a country has experienced economic development?

- ☒ A an improvement in the average citizen's quality of life
- B an increase in the country's real GDP
- C an improvement in the country's trade balance
- D an appreciation in the country's currency, results in trade deficit, incomes ↓

} caused economic growth

Economic Development

The process in which people in a country become wealthier, healthier, better educated, and have greater access to good quality housing.



Economic development is the process in which a nation is being improved in the sector of the economic, political, and social well-being of its people.

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26 Which combination of factors is most likely to result in more rapid economic growth?

- A increases in employment and in the balance of payments deficit *one +ve, one -ve*
Labour force
 B increases in the level of investment and in the size of the working population *↑ Multiplier*
 C more equal distribution of wealth and a higher level of unemployment benefits *Dependency on government*
 D more rapid inflation and an increase in the national debt *Negative only*

27 How might a developing economy gain from a multilateral reduction in import tariffs and the removal by developed economies of subsidies on food exports? *Export prices ↓ Imports expensive*

- A through increased specialisation leading to higher productivity
 B through increased ability to protect infant industries
 C through a reduction in the cost to the economy of imported food
 D through increased tariff revenues

28 An economy with a floating exchange rate is in recession and at the same time has a deficit on the current account of its balance of payments. *Incomes ↓*

Which policy combination would be most likely to help with both of these problems?

	interest rates	tax rates
A	decrease ✓	unchanged ✓
B	decrease ✓	increase →
C	increase	unchanged ✓
D	increase	increase →

save ↑ spend ↓ C ↓ AD ↓ incomes ↓

*↓ save ↓ spend ↑ C ↑ AD ↑ growth
 ↓ Depreciation, X prices ↓, B of T surplus
 M prices ↑*

[interest rate and exchange rate have a positive relationship]

29 In 2004 China's ability to exploit its comparative advantage in cotton production increased.

What could explain this change?

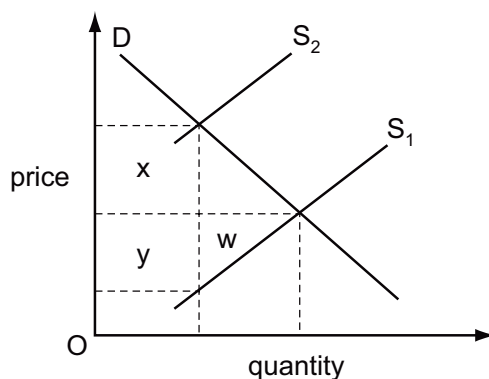
Low opportunity cost, Quality ↑

- A a fall in the value of the currency of India, a major cotton producer
 B a reduction of the import quota on Chinese cotton into the European Union *This means restricting imports*
 C a rise in the wages of Brazilian cotton workers matched by an increase in their productivity
 D the removal of the United States of America's subsidy to its cotton growers

Now China's goods are more competitive

Removal of subsidy, less protectionism, more free trade, Specialization

30 In the diagram D is the demand curve for Indian tea exports and S_1 is the initial supply curve.

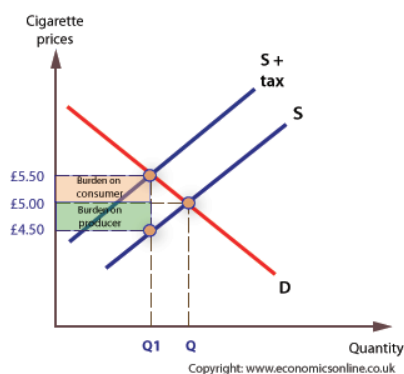


The Indian government imposes a tax on tea exports, which causes the supply curve to shift to S_2 .

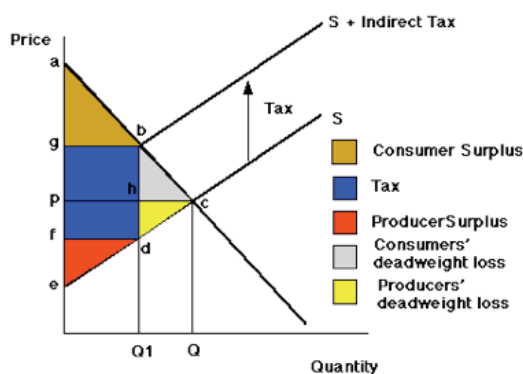
$$C \cdot B + P \cdot B = \text{Tax revenue}$$

Which areas in the diagram measure the resulting gain in tax revenue to the Indian government and the resulting loss in producer surplus to its tea producers?

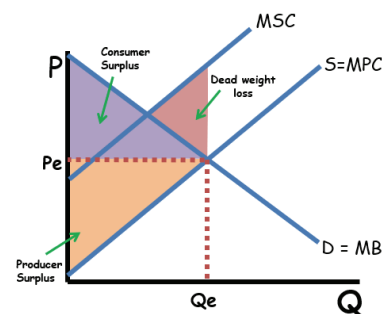
	gain in tax revenue	loss in producer surplus
A	x	w + y
B	x + y	y
C	x	y
D	x + y	w + y



$$C \cdot B + P \cdot B \Rightarrow \text{Tax revenue}$$



Producer surplus after tax



Producer surplus before tax

difference is loss of producer surplus

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